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Comenius University Bratislava
Faculty of Mathematics, Physics and Informatics

THESIS ASSIGNMENT

Name and Surname:	Marek Šugár
Study programme:	Data Science (Joint degree study, bachelor I. deg., full time form)
Field of Study:	Computer Science Mathematics
Type of Thesis:	Bachelor´s thesis
Language of Thesis:	English
Secondary language:	Slovak

Title: Analysis of financial markets using machine learning

Annotation: Modeling and prediction of stocks in financial markets have long been a researched problem. Historically, this problem was approached using specific methods of financial mathematics, but with the significant development and success of machine learning in general, machine learning methods have also begun to be applied to predicting the behavior of financial markets. The aim of this thesis is to create a prediction model for selected stocks, which involves the appropriate selection and identification of relevant explanatory variables, as well as the suitable choice of model. Subsequently, it will be possible to analyze whether the obtained predictions can be used for successful creation of an investment portfolio.

Supervisor:	Mgr. Samuel Rosa, PhD.
Department:	FMFI.KAMŠ - Department of Applied Mathematics and Statistics
Head of department:	doc. Mgr. Igor Melicherčík, PhD.
Assigned:	02.10.2025
Approved:	28.10.2025

doc. Mgr. Tomáš Vinař, PhD.
Guarantor of Study Programme

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Student

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Supervisor